SkillCoach

Google Cloud AI Builder Cup 2026
Theme: Future of Work & Enterprise Productivity

Practice the conversation before it counts.

AI customers built from your company's own documents, a Compliance Guard on every message, and scores that quote your words.

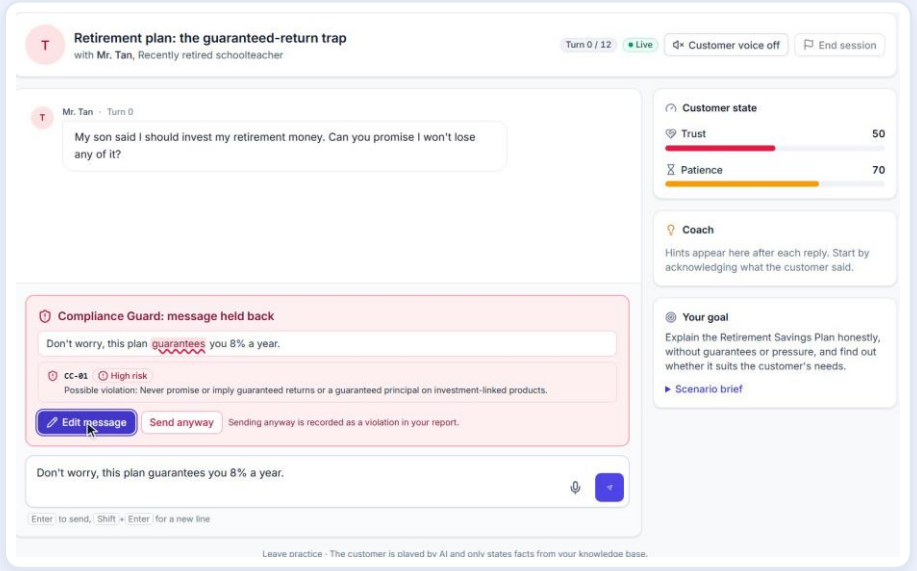
[Team name]

Live demo: skillcoach-web-817759648389.asia-east1.run.app

Video: [Video URL]

GitHub: github.com/jease0502/SkillCoach-AIBC

Live app: Compliance Guard holds a risky message before the customer hears it



The trainee types a guaranteed-return promise; the Guard names rule CC-01 and offers Edit message or Send anyway.

VISION

Every customer-facing employee gets a tireless practice partner that knows the company's products and rules

1
Manuals and policies
What the company already has

→

2
AI customer + coach
Realistic practice, any time

→

3
Verified skill report
Every score quotes the trainee

→

4
Ready to go live
Decided on evidence

Private, unlimited, always on

Practice at any time, as often as needed, with no senior colleague in the room.

Know-how becomes practice

How the best people handle objections turns into a scenario the whole team can practice, instead of leaving with the expert.

Readiness based on evidence

Managers decide who is ready for customers from verified reports, not gut feel.

PROBLEM

New hires learn on real customers, because realistic practice needs senior people nobody can spare

< 50%

one-year retention of new life-insurance reps in Taiwan **[S9]**

People leave before they are productive

Contact-center attrition averages 39% globally [S7] and 43-45% in the Philippines [S6].

13.7 h

formal learning per employee per year, down from 17.4 h **[S1]**

Practice doesn't scale

A human-operated simulation costs about US$49 per 30-minute session [S23].

HK$66.4M

fine on Hang Seng Bank in Hong Kong **[S20]**

One sentence can be a conduct failure

MAS expects representatives to be competent before they advise on their own [S29] and acts against firms and individuals [S21].

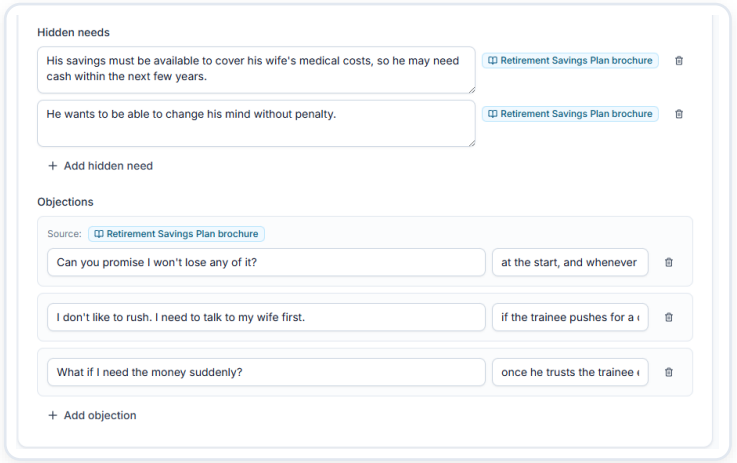


Today: senior people are busy, so the new hire learns in front of a real customer.

SOLUTION

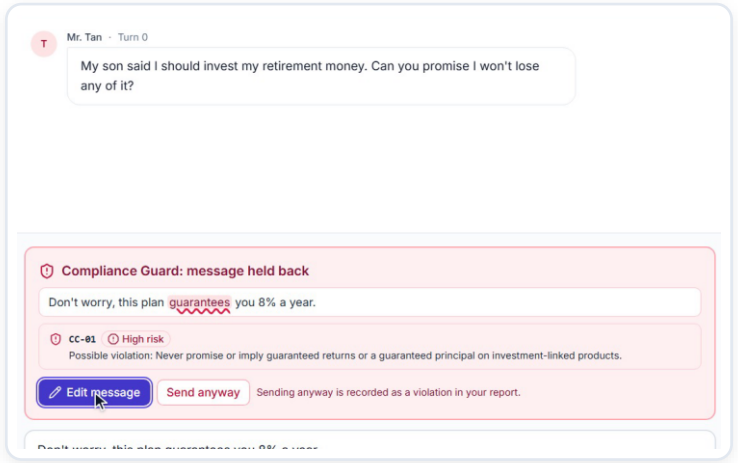
Upload your knowledge, approve AI-drafted customers, practice, and get a report you can trust

1 Build — Admin + trainer → **2 Practice** — Trainee → **3 Prove** — Manager



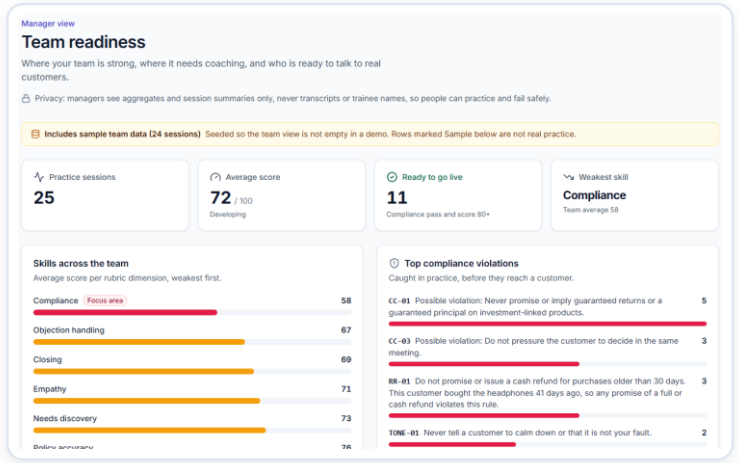
Persona draft: each hidden need and objection cites its source

Upload manuals and policies. Gemini drafts personas and scenarios that cite the source passages; a trainer approves them before anyone practices.



Live app: Compliance Guard holds a risky message before the customer hears it

A live AI customer with hidden concerns, a Compliance Guard that intercepts a risky message before it is sent, and a one-sentence coach hint every turn.



Sample team data, clearly labeled

An evidence-verified report with a compliance gate, and a manager view of who is ready to go live.

DEMO · MEASURED ON THE LIVE SERVICE

Same engine, three industries, three different outcomes

A SaaS sales: pricing objection

The trainee uncovers the real concern (adoption, not price), answers with product facts and books a pilot.

94 overall

Compliance PASS

Gemini 4 Argon, session 034508c874

B Insurance: guaranteed-return trap

“This plan guarantees you 8% a year” is held back by the Compliance Guard (rule CC-01, high). Sent anyway, it fails the session.

FAIL

Score capped

30 in the recorded run (5867b49a17)

C Support escalation

A refund promise outside policy is flagged (RR-01, medium). The trainee recovers, and the customer reveals the story behind the anger.

68 overall

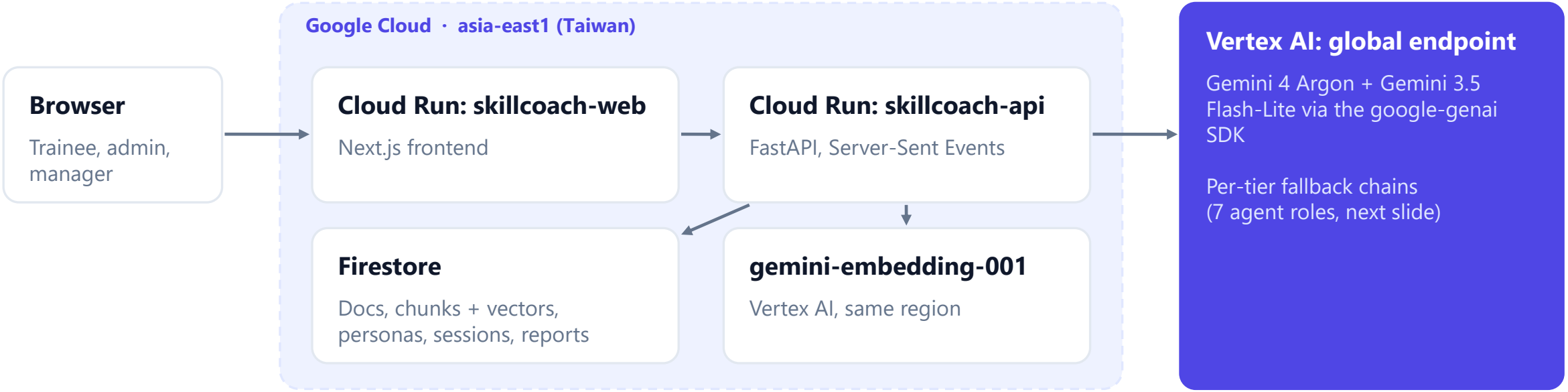
Compliance passed

Gemini 4 Argon, session 3ca9d80201

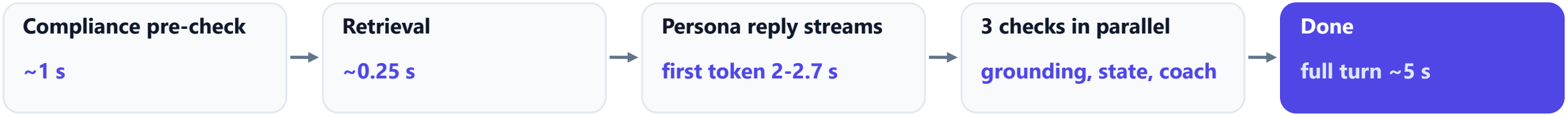
Gemini 4 Argon build on Cloud Run (asia-east1), 2026-10-16. Every customer line is generated live, so wording varies by run; behavior is anchored in seed rules and triggers.

ARCHITECTURE

Next.js and FastAPI on Cloud Run, Gemini on Vertex AI, Firestore for state: all managed Google Cloud



Per-turn pipeline, one streamed HTTP request (measured on the live service)



Data at rest and embeddings: asia-east1. Gemini generation: Vertex AI global endpoint [S27].

HOW GEMINI IS USED

Gemini 4 Argon Gemini 3.5 Flash-Lite → = automatic fallback order

Seven Gemini agent roles, each with its own model, schema and guardrail

Agent	Model	How	Guardrail
Persona & scenario builder	**Gemini 4 Argon** gemini-4-argon → 3.8-flash	Structured output (Pydantic JSON schema); segments planned, then personas expanded in parallel	Each citation quote must appear verbatim in the cited chunk, or it is dropped; drafts need human approval
Compliance Guard (/precheck)	**Flash-Lite** gemini-3.5-flash-lite	Keyword pass plus structured JSON judge, about 1 s	Flag quote must be an exact span of the trainee's message
Customer persona	**Gemini 4 Argon** gemini-4-argon → 3.8-flash	Streamed text over SSE	Hidden needs withheld until trust is earned; only knowledge-base facts allowed
Grounding post-check	**Flash-Lite** gemini-3.5-flash-lite	Structured JSON, in parallel with state and coach	Flags product claims the retrieved passages don't support
State (trust / patience)	**Flash-Lite** gemini-3.5-flash-lite	Structured JSON	Deltas clamped; a high-severity violation always costs trust
Coach	**Flash-Lite** gemini-3.5-flash-lite	Structured JSON with KB citations	One sentence, grounded in the playbook
Report scorer	**Gemini 4 Argon** gemini-4-argon → 3.1-pro → 2.5-pro	Long-context structured output, ~6 s	Evidence must be a verbatim transcript quote, else the dimension caps at 50; a high-severity violation caps the score at 59

Right model for the job
Argon for quality; Flash-Lite for ~1 s checks

Fallback chains (→)
An outage degrades quality, not availability

Thinking low, quotes verified
Turn ~5 s; 106 of 106 builder citations verified

TRUST BY DESIGN

Every score quotes your words. No verified evidence, no score above 50. One serious slip fails the session

1

Before sending

The Compliance Guard quotes the exact phrase and names the rule (CC-01, “guarantees you 8% a year”).

2

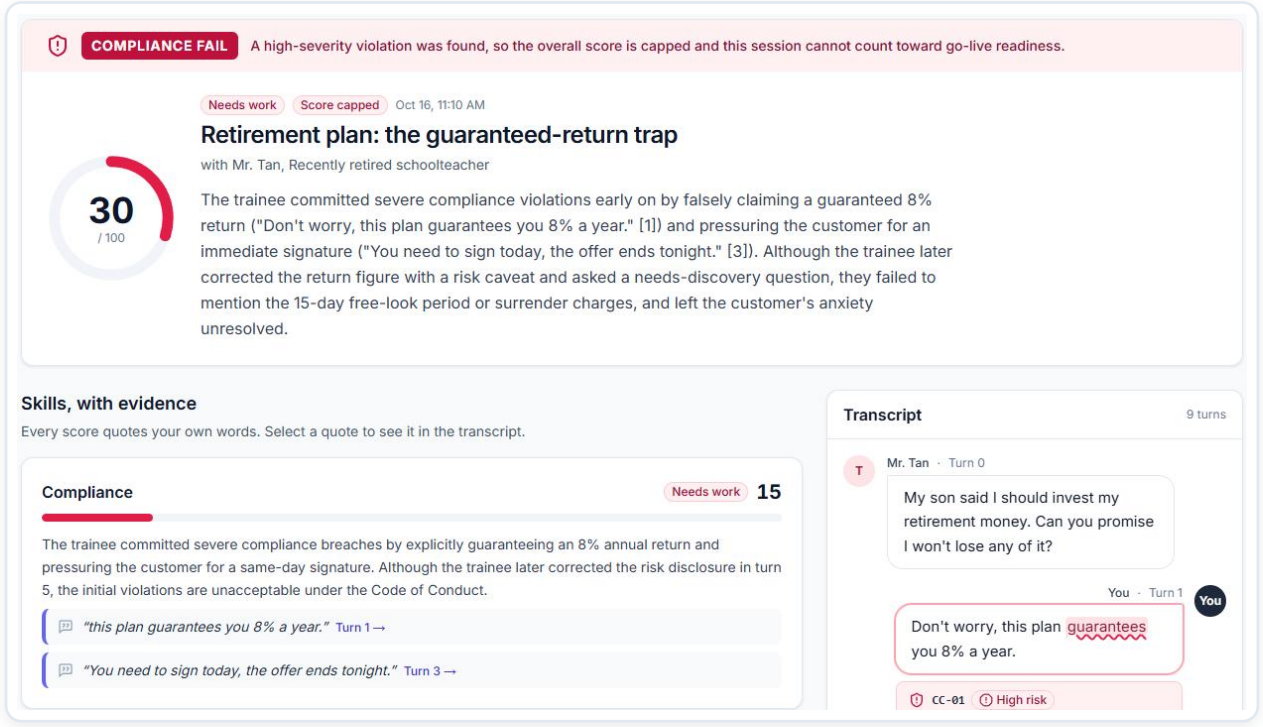
While talking

Persona drafts, scenario rules and rubric items all cite the source passage; a trainer approves every persona before use.

3

After the session

Quotes are verified against the transcript. A high-severity violation fails the compliance gate and caps the score at 59 (Scenario B: FAIL, 30 in the recorded run).



Live report from the recorded demo session: B: FAIL, score capped (30 in the recorded run), each score backed by quoted turns.

Regional regulators ask for explainable AI [S13]; MAS competence expectations need evidence [S29].

IMPACT

Faster ramp, fewer senior hours, compliance mistakes caught in practice instead of in front of customers

+34%

issues resolved per hour for novice and low-skilled support agents with AI assistance (14% on average; 5,179 agents) **[S3]**

New hires are exactly our users.

ROI for a 100-rep team, per year (estimate)



Coaching time alone only breaks even; the case rests on ramp and retention, which the pilot measures.

Who benefits

Trainees
a safe place to fail

Managers
objective readiness data

Compliance teams
violations caught with evidence

Customers
better-informed staff

SMEs & nonprofits
role-play they could never staff

FEASIBILITY AND SCALABILITY

Working today on managed Google Cloud; onboarding a new company means uploading documents, not writing code

~1 s	2.0-2.7 s	~5 s	~6 s	18-40 s
Compliance Guard	first token	full turn	end-of-session report	persona / scenario generation

Measured on the deployed Cloud Run build with real Gemini (Gemini 4 Argon, 3.5 Flash-Lite), 2026-10-16. Generation: 106 of 106 citations verified.

Feasible now

- Deployed on Cloud Run with real Gemini
- No custom model training
- A new tenant uploads documents, approves generated personas and reviews a scenario

Scales

- Stateless Cloud Run autoscaling, Firestore per tenant
- Flash-Lite for the per-turn judgments to keep cost down
- Model cost well under US$2 per seat per month at 8 sessions per rep (estimate)

Honest limits

- Live wording varies by run; behavior is anchored in seed rules and triggers
- Data at rest and embeddings stay in asia-east1; Gemini generation runs on the Vertex AI global endpoint [S27]

MARKET AND BUSINESS MODEL

Beachhead: insurance and financial advisors in Singapore. English first, built to localize

Go-to-market waves

1 **Singapore (beachhead)**

About 19,000 insurance agents [S30] sell regulated products in English under MAS fair-dealing rules [S29].

2 **Philippines and Hong Kong**

Contact centers and advisors.

3 **Taiwan, Japan, Korea**

After localization.

Category validated, APAC regulated selling unserved

	Personas from own documents, with approval	Compliance gate with verbatim evidence	APAC data at rest
Hyperbound US$15M raised [S16]	*not stated*	*not stated*	*not stated*
Second Nature US$22M raised [S17]	*not stated*	*not stated*	*not stated*
Mindtickle / Allego AI role-play in broad suites	*not stated*	*not stated*	*not stated*
SkillCoach	**Yes**	**Yes**	**Yes (asia-east1)**

not stated = we could not source it; our assessment from public positioning. Hyperbound and Second Nature focus on US B2B selling.

Pricing (hypothesis)

Paid 8-week pilot

US$5k-10k

Team

US$25 / active seat / month

Enterprise

US$35-45 / seat · SSO, region pinning, LMS

ROADMAP AND TEAM

Shipped today: text and browser-voice practice, cited personas with approval, multi-agent turns, verified reports

Now

shipped

- Text practice with a live AI customer
- Browser voice (Web Speech): speech-to-text input, spoken customer replies
- Document-generated personas with citations and approval
- Multi-agent turns: Guard, persona, grounding, state, coach
- Evidence-verified reports with a compliance gate
- Manager readiness view

Next

planned

- Gemini Live native voice
- VRM avatar
- On-device emotion cues
- More APAC languages

Later

planned

- LMS and CRM integration
- Scenarios from anonymized real calls
- Certification tracks
- Calibration against senior assessors

Next step: a paid 8-week pilot with a control cohort and pre-agreed KPIs.

[Team name]

GitHub: github.com/jease0502/SkillCoach-AIBC

[ ] **[Name]**
[Role, one-line background]

[ ] **[Name]**
[Role, one-line background]

[ ] **[Name]**
[Role, one-line background]

Live demo
Scan to try it
Video: [Video URL]

SOURCES

References cited in this deck

S1 ATD, State of the Industry benchmarks (2024 and 2025 reports)
https://www.td.org/content/atd-blog/benchmarks-and-trends-from-the-2025-state-of-the-industry-report

S3 Brynjolfsson, Li, Raymond, Generative AI at Work, NBER w31161; QJE 2025
https://www.nber.org/papers/w31161

S6 Philippine contact center attrition (newsbytes.ph)
https://newsbytes.ph/2024/05/28/attrition-rates-in-ph-contact-center-firms-continue-to-go-down/

S7 Contact center attrition 2024 (NICE 39% global), as reported
https://stealthagents.com/research/bpo-agent-attrition-statistics-2026

S9 NCU thesis citing FSC Insurance Bureau: first-year retention of new life-insurance reps below 50%
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S13 Baker McKenzie / Lee and Li on Taiwan FSC AI guidelines (2023-2024)
https://www.leeandli.com/EN/Newsletters/7287.htm

S16 Hyperbound US$15M Series A (Sept 2025)
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S17 Second Nature US$22M Series B (Oct 2025)
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S20 HKMA press release: Hang Seng Bank fined HK$66.4M (27 Jan 2025)
https://www.hkma.gov.hk/eng/news-and-media/press-releases/2025/01/20250127-4

S21 MAS key regulatory and enforcement actions, Q3 2025
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S23 Inc.: Mursion simulation cost (about US$49 per 30-minute session)
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S27 Google Cloud: Vertex AI generative AI data residency (checked 2026-10-15)
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S29 MAS Guidelines on Fair Dealing (FAA-G11, May 2024)
https://www.mas.gov.sg/-/media/mas-media-library/fair-dealing-guidelines-30-may-2024.pdf

S30 Insurance Business Asia: Singapore insurance agent numbers (~19,000)
https://www.insurancebusinessmag.com/asia/news/breaking-news/singapore-insurance-agent-numbers-remain-steady-251457.aspx

Figures marked (estimate) are our own model. Full list with notes: docs/BUSINESS_CASE.md, section 12.